Learning Architecture

AI Literacy for the Modern Workforce

This is the learning architecture behind *AI Literacy for the Modern Workforce*: four modules on a Context → Evidence → Mechanism → Application arc, each mapped to the 4D competencies, the practice activities that build them, the performance objectives they satisfy, and the assessment tier that measures each one. The whole structure was designed to feed a Kirkpatrick evaluation: every objective resolves to a tier, and two objectives produce artifacts the 30/60/90-day behavioral follow-up can track.

The program was scoped for two delivery platforms in parallel: a custom-coded React environment (Component 4, the platform you are reading this on) and an Articulate Rise/Storyline build (Component 5). The custom platform shipped. The Articulate build was deferred as a budget decision, not a design one, and the resource plan gives the rationale. The architecture keeps both columns so the instructional design stays platform-agnostic: only the interactive components are platform-dependent, and the content could be rebuilt in any equivalent authoring tool from the same objectives and activities.

Module 3 also inherits four design principles from the capstone pilot that preceded this build (Ritchot, 2025, pp. 44–45). Scaffold abstract content from concrete analogies upward. Visualize the invisible with diagrams and tokenizer demos. Embed reflection and scenario-based application. Chunk content, with feedback after each part. Three Module 3 components answer that pilot's specific findings. The tokenizer playground (P5) is the pilot's first recommendation and answers its participants' request for a tokenizer sandbox (pp. 37, 41, 43). The number-tokenization comparison in §3.2 descends from one of the two visuals participants named as making the content concrete (p. 43). The playground's free mode shows the token count and compression ratio for any text, so learners can test the first half of the pilot's cheat-sheet rule, clean input means fewer tokens and better output, which every pilot participant applied in the scenario task (pp. 38, 41).

Reading the diagram

- Click any module card to reveal its dual-platform delivery, its practice activities with interaction types, and its performance objectives with competency tags,

assessment tier, and evaluation mapping.

- Expand the Kirkpatrick lane to read what each evaluation level measures. Objectives tagged → **L3** feed the 30/60/90-day behavioral plan.

4 MODULES **17** OBJECTIVES **12** PRACTICE ACTIVITIES **2** PLATFORMS **8** COMPETENCY-ASSESSED **2** L3 TRACKED

CUSTOM PLATFORM · COMPONENT 4

React-based learning environment. Interactive dashboards, tokenizer playground, AI interaction sandbox, learning analytics. Primary technical differentiator.

ARTICULATE PLATFORM · COMPONENT 5

Rise 360 + Storyline. Responsive scroll-based content, branching scenarios with consequence-based feedback, SCORM/xAPI assessments. Ecosystem fluency proof.

LEARNING PROGRESSION — *Context → Evidence → Mechanism → Application*

1 CONTEXT

Why AI Literacy Matters Now

Strategic context, market forces, labor market transformation

35–45 min | DELEGATION DILIGENCE | 2 activities · 4 objectives

4C · CUSTOM PLATFORM

Interactive Explorations

Interactive data narratives: scrollable data stories combining WEF and Anthropic findings with embedded reflection prompts

5B · ARTICULATE PLATFORM

Rise 360

Responsive scroll-based content with embedded interactions and scenario-based knowledge checks

PRACTICE ACTIVITIES (2)

- **P1** Interactive Data Narrative — *Guided data exploration*
- **P2** Stigma Data Reflection — *Reflective prompt (private)*

PERFORMANCE OBJECTIVES (4)

- **1.1** Interpret adoption data; identify implications for own role — DELEGATION ○ T1 Knowledge check
- **1.2** Articulate business case using three quantified findings — DELEGATION ○ T1 Knowledge check
- **1.3** Explain 69% concealment dynamic and shared vocabulary solution — DILIGENCE ○ T1 Knowledge check
- **1.4** Locate own usage on augmentation-automation spectrum — DELEGATION ◐ T2 Practice activity (P1)

2 How AI Is Actually Being Used at Work

EVIDENCE

Real usage patterns, productivity data, behavioral reality vs. hype

45–60 min | DELEGATION DISCERNMENT | 2 activities · 4 objectives

4A · CUSTOM PLATFORM

Live Data Dashboard

Filterable dashboard covering augmentation/automation split, occupation-level adoption data, collaboration patterns, productivity distributions, and guided reflection prompts

5B · ARTICULATE PLATFORM

Rise 360

Responsive scroll-based content with embedded data visualizations and scenario-based knowledge checks

PRACTICE ACTIVITIES (2)

- **P3** Dashboard: Augmentation vs. Automation — *Filterable dashboard*
- **P4** Dashboard: Productivity Distributions — *Filterable dashboard*

PERFORMANCE OBJECTIVES (4)

- **2.1** Distinguish augmentation from automation against behavioral data — DELEGATION · T2 · Practice activity (P3)
- **2.2** Analyze task-level productivity data for high-gain vs. high-risk — DELEGATION DISCERNMENT · T1 · Knowledge check
- **2.3** Identify one task to add AI, one to increase oversight — DELEGATION DISCERNMENT · T2 · Practice activity (P4) → Level 3 → L3
- **2.4** Describe self-report/behavioral divergence and directive patterns — DELEGATION · T1 · Knowledge check

3 Understanding How Language Models Work

MECHANISM

Tokenization, attention, probability-based generation, hallucination mechanics

50–65 min | DESCRIPTION DISCERNMENT | 3 activities · 4 objectives

4C · CUSTOM PLATFORM

Interactive Explorations

Tokenizer playground, next-token prediction demonstration with parameter adjustment, context window visualization

5A · ARTICULATE PLATFORM

Storyline Branching

Decision simulator scenarios with consequence-based feedback on model behavior interpretation

PRACTICE ACTIVITIES (3)

- **P5** Tokenizer Playground — *Input-output sandbox*
- **P6** Next-Token Prediction Demo — *Parameter-adjustment interactive*
- **P7** Context Window Scenario

PERFORMANCE OBJECTIVES (4)

- **3.1** Explain tokenization; predict unexpected results; describe systematic errors — DELEGATION DISCERNMENT · T3 · Practice (P5) + Competency
- **3.2** Describe next-token prediction vs. retrieval; explain fluent fabrication — DELEGATION DISCERNMENT · T3 · Practice (P6) + Competency
- **3.3** Identify three systematically unreliable task categories

Simulated failure scenario

DELEGATION DISCERNMENT ● T3 Knowledge check + Competency

3.4 Determine whether output is within reliable capability range

DISCERNMENT ● T3 Practice (P5–P7) + Competency

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4 APPLICATION

Evaluating AI Outputs and Working Responsibly

Applied judgment, output verification, responsible delegation

70–105 min | DELEGATION DESCRIPTION DISCERNMENT DILIGENCE | 5 activities · 5 objectives

4B · CUSTOM PLATFORM

AI Interaction Sandbox

Structured roleplay, feedback simulator, prompt reformulation workshop, annotation layer, diligence statement builder

5A · ARTICULATE PLATFORM

Storyline Branching

Multi-path branching scenarios with compounding consequences and constructed response assessment

PRACTICE ACTIVITIES (5)

P8 Task Decomposition Exercise
Categorization with consequences

P9 Prompt Reformulation Comparison
Before/after workshop

P10 Output Verification Scenario
Sequential element classification

P11 Iterative Refinement Exercise
Multi-turn guided interaction

P12 Diligence Statement Practice
Constructed response

PERFORMANCE OBJECTIVES (5)

4.1 Decompose task into human-only, AI-assisted, and delegated components
DELEGATION ● T3 Practice (P8) + Competency

4.2 Reformulate underspecified prompt into structured request
DESCRIPTION ● T3 Practice (P9) + Competency

4.3 Evaluate AI deliverable: verify, detect hallucination, classify elements
DISCERNMENT DILIGENCE ● T3 Practice (P10) + Competency

4.4 Iterate through 3+ refinement turns using Description-Discernment loop
DESCRIPTION DISCERNMENT ◐ T2 Practice (P11)

4.5 Produce AI diligence statement using 4D vocabulary
DILIGENCE ● T3 Practice (P12) + Competency + Level 3 → L3

ASSESSMENT ARCHITECTURE — *Feeds Kirkpatrick Levels 2–4 below*

○ **Tier 1: Knowledge checks**

Scenario-based knowledge checks with explanatory feedback, embedded within modules

5 objectives

◐ **Tier 2: Practice**

Learner choices and outputs within interactives, sandbox, and dashboards, assessed in-flow

4 objectives

● **Tier 3: Competency and transfer**

Competency assessment (5C) and Level 3 behavioral follow-up (6C): post-program transfer measurement

8 objectives

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KIRKPATRICK EVALUATION FRAMEWORK

L1	L2	L3	L4
Reaction	**Learning**	**Behavior**	**Results**
Level 1	Level 2	Level 3	Level 4

L1 **Reaction**

Post-program survey across four actionable dimensions: perceived relevance, confidence change, content quality, and intent to apply, plus a standard NPS item

L2 **Learning**

Twelve-item scenario-based pre/post assessment, parallel-form to resist gaming, measuring knowledge gain across usage patterns, failure modes, mechanics, and evaluation

L3 **Behavior**

Manager evidence review and participant self-assessment at 30/60/90 days, rating twelve 4D-mapped behavioral indicators; objectives 2.3 and 4.5 produce the trackable artifacts

L4 **Results**

Business outcomes downstream of competent use: efficiency (time-to-first-draft, revision cycles), quality (AI-error rate), transparency, and time-to-competency, run through a Phillips ROI model (156% in the illustrative worked example)

L3 **Transfer Measurement by Design**

Two objectives produce trackable artifacts that the 30/60/90-day behavioral follow-up checks for sustained use. **Objective 2.3** is a task-level action commitment: the participant identifies a specific task they will change their approach to. **Objective 4.5** is an AI diligence statement: the participant documents AI's role in a completed work product using 4D vocabulary. Level 3 data shows whether the program produces behavior change.

ANALYTICS INFRASTRUCTURE — *Hidden from the learner; supports Levels 2 & 3*

4D · ANALYTICS DASHBOARD

Per-learner progress tracking, module completion rates, practice activity engagement metrics, and time-on-task data.

Portfolio-mode analytics dashboard at /#/dashboard (Ctrl/Cmd+Shift+A enters portfolio mode). Surfaces the practice-activity and KC data that feeds Kirkpatrick Levels 2 and 3.

4E · PLATFORM SHELL

Sidebar navigation, the 90% scroll-completion sentinel, hash router, theme system, and the learner-progress and analytics contexts.

Wraps every module: every section renders inside the shell's SectionContainer and consumes its shared progress and analytics providers.

PRACTICE ACTIVITY DISTRIBUTION



Practice density shifts from context-setting in Modules 1–2 to applied skill-building in Modules 3–4. Of the 12 activities, 10 involve hands-on interaction.

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